How to transfer data between XML and MySQL

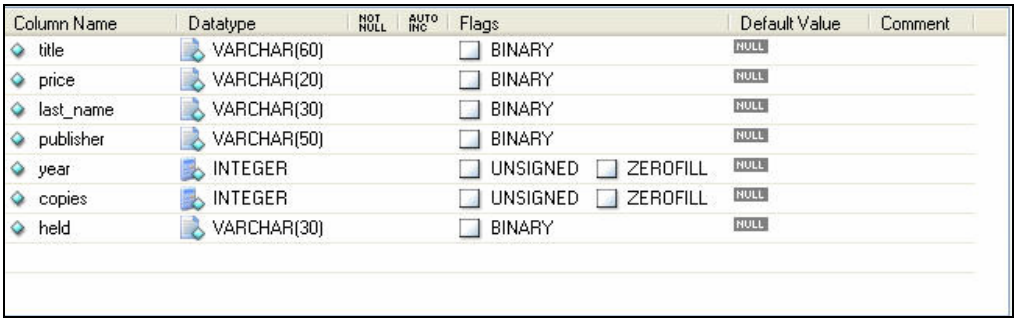
Andy Tang

1. Importing XML data into MySQL database.

1.1. The first, a table named 'books' was created, with field name and number corresponding with the ones in XML data file.

```
create table books
(
title varchar(60),
price varchar(20),
last_name varchar(30),
publisher varchar(50),
year int,
copies int,
held varchar(30)
);
```

On executing the "create table" command, the result of table structure is demonstrated as below.

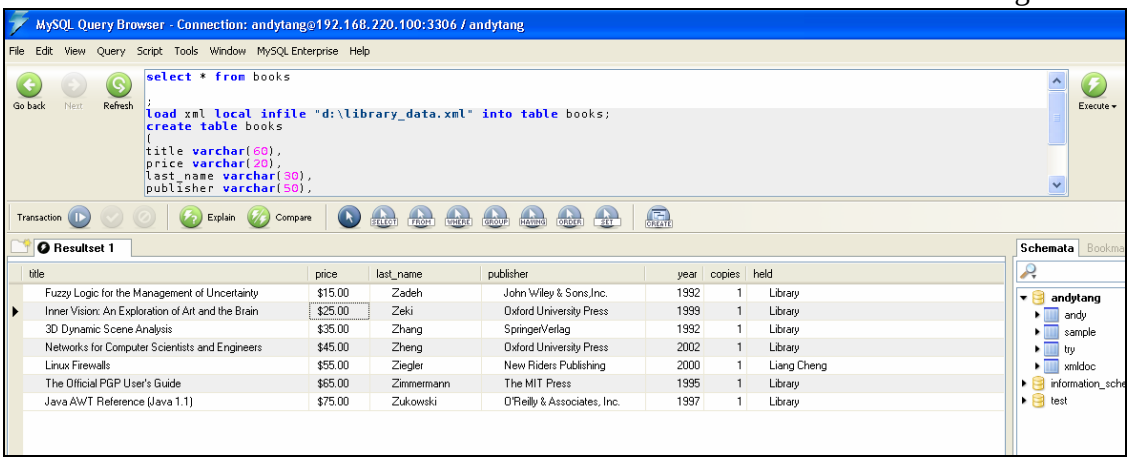


Column Name	Datatype	NOT NULL	AUTO INC	Flags	Default Value	Comment
title	VARCHAR(60)			BINARY	NULL	
price	VARCHAR(20)			BINARY	NULL	
last_name	VARCHAR(30)			BINARY	NULL	
publisher	VARCHAR(50)			BINARY	NULL	
year	INTEGER			UNSIGNED ZEROFILL	NULL	
copies	INTEGER			UNSIGNED ZEROFILL	NULL	
held	VARCHAR(30)			BINARY	NULL	

1.2. The second, a SQL query named 'LOAD' is employed to ensure that the data contained in XML is accurately imported into MySQL table built in prior time.

Query employed: "load xml local infile 'd:\library_data.xml' into table books;"

After executing 'select * from books', the contents in XML file is parsed and transferred into table which we build before..



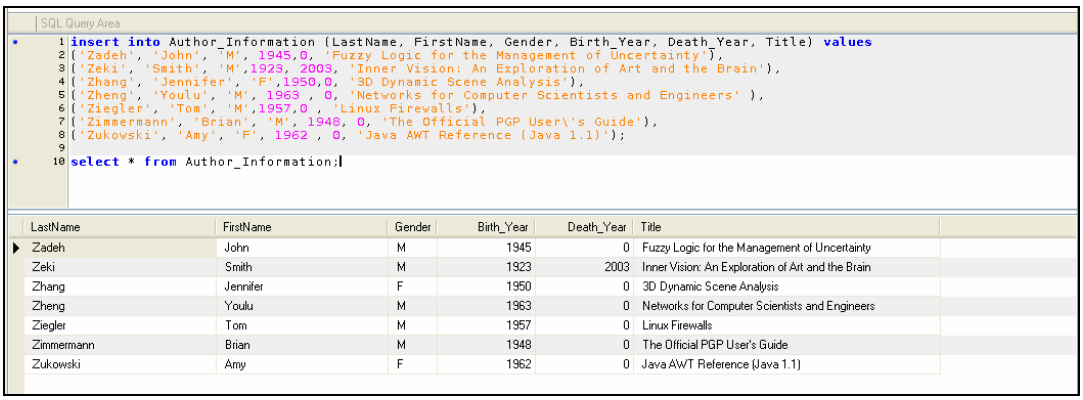
2. Creating a table in MySQL and export it as a XML file format.

2.1. Creating table

```
create table Author_Information
(
LastName varchar(30) null,
FirstName Varchar(30) null,
Gender Varchar(1) null,
Birth_Year int null,
Death_Year int null,
Title Varchar(60) null
);
```

2.2. Inject records into table.

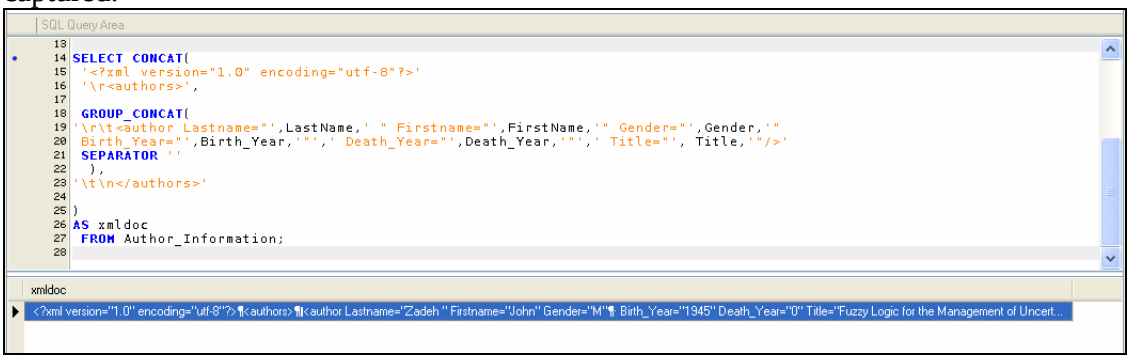
```
insert into Author_Information (LastName, FirstName, Gender, Birth_Year, Death_Year, Title) values
('Zadeh', 'John', 'M', 1945,0, 'Fuzzy Logic for the Management of Uncertainty'),
('Zeki', 'Smith', 'M',1923, 2003, 'Inner Vision: An Exploration of Art and the Brain'),
('Zhang', 'Jennifer', 'F',1950,0, '3D Dynamic Scene Analysis'),
('Zheng', 'Youlu', 'M', 1963 , 0, 'Networks for Computer Scientists and Engineers' ),
('Ziegler', 'Tom', 'M',1957,0 , 'Linux Firewalls'),
('Zimmermann', 'Brian', 'M', 1948, 0, 'The Official PGP User\'s Guide'),
('Zukowski', 'Amy', 'F', 1962 , 0, 'Java AWT Reference (Java 1.1)');
```



2.3. Export table contents into a XML file

```
SELECT CONCAT(
 '<?xml version="1.0" encoding="utf-8"?>'
 '\r<authors>',
        GROUP_CONCAT(
                '\r\t<author Lastname="', LastName, '" Firstname="', FirstName,  '"
                Gender="', Gender,   '" Birth_Year="', Birth_Year, '",   ', Death_Year="',
                Death_Year, '",    ', Title="', Title,  '"/>'
                 SEPARATOR ''
                            ),
                '\t\n</authors>'
                    )
AS xmldoc
 FROM Author_Information;
```

After execution of the “select” query, the xml contents are generated and displayed as below. In event of making demonstration clear, the code screen with color mark is captured.



The generated XML code is listed as below.

```
<?xml version="1.0" encoding="utf-8"?>
<authors>
        <author Lastname="Zadeh " Firstname="John" Gender="M"
 Birth_Year="1945" Death_Year="0" Title="Fuzzy Logic for the Management of Uncertainty"/>
        <author Lastname="Zeki " Firstname="Smith" Gender="M"
 Birth_Year="1923" Death_Year="2003" Title="Inner Vision: An Exploration of Art and the Brain"/>
        <author Lastname="Zhang " Firstname="Jennifer" Gender="F"
 Birth_Year="1950" Death_Year="0" Title="3D Dynamic Scene Analysis"/>
        <author Lastname="Zheng " Firstname="Youlu" Gender="M"
 Birth_Year="1963" Death_Year="0" Title="Networks for Computer Scientists and Engineers"/>
        <author Lastname="Ziegler " Firstname="Tom" Gender="M"
 Birth_Year="1957" Death_Year="0" Title="Linux Firewalls"/>
        <author Lastname="Zimmermann " Firstname="Brian" Gender="M"
 Birth_Year="1948" Death_Year="0" Title="The Official PGP User's Guide"/>
        <author Lastname="Zukowski " Firstname="Amy" Gender="F"
 Birth_Year="1962" Death_Year="0" Title="Java AWT Reference (Java 1.1)"/>
</authors>
```

In order to make the code easily read, the generated XML file is opened by use of Visual Studio, and the screen is captured and demonstrated below.

authordata.xslt authordata.xml

```
<?xml version="1.0" encoding="utf-8"?>
<?xml-stylesheet type="text/xsl" href="authordata.xslt"?>
<authors>
    <author Lastname="Zadeh " Firstname="John" Gender="M"
Birth_Year="1945" Death_Year="0" Title="Fuzzy Logic for the Management of Uncertainty"/>
    <author Lastname="Zeki " Firstname="Smith" Gender="M"
Birth_Year="1923" Death_Year="2003" Title="Inner Vision: An Exploration of Art and the Brain"/>
    <author Lastname="Zhang " Firstname="Jennifer" Gender="F"
Birth_Year="1950" Death_Year="0" Title="3D Dynamic Scene Analysis"/>
    <author Lastname="Zheng " Firstname="Youlu" Gender="M"
Birth_Year="1963" Death_Year="0" Title="Networks for Computer Scientists and Engineers"/>
    <author Lastname="Ziegler " Firstname="Tom" Gender="M"
Birth_Year="1957" Death_Year="0" Title="Linux Firewalls"/>
    <author Lastname="Zimmermann " Firstname="Brian" Gender="M"
Birth_Year="1948" Death_Year="0" Title="The Official PGP User's Guide"/>
    <author Lastname="Zukowski " Firstname="Amy" Gender="F"
Birth_Year="1962" Death_Year="0" Title="Java AWT Reference (Java 1.1)"/>
</authors>
```

3. Create XSLT to display the XML file.

In case of making author death year in optional display, a “XML:choose” syntax is employed to ensure to display related information in flexibility. The screen capture of XSLT with color code is demonstrated below.

```
<?xml version="1.0" encoding="utf-8"?>
<xsl:stylesheet version="1.0"
    xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
<xsl:template match="authors">
    <html>
    <body>
            <TABLE BORDER="1" CELLPADDING="5">
                <THEAD>
                    <TH>Last Name</TH>
                    <TH>First Name</TH>
                    <TH>Gender</TH>
                    <TH>Birth Year</TH>
                    <TH>Death Year</TH>
                    <TH>Title</TH>
                </THEAD>
                <xsl:for-each select="author" >
                        <xsl:sort  select="author" order="ascending"/>
                    <TR ALIGN="CENTER">
                        <TD>
                            <xsl:value-of select="@Lastname"/>
                        </TD>
                        <TD>
                            <xsl:value-of select="@Firstname"/>
                        </TD>
                        <TD>
                            <xsl:value-of select="@Gender"/>
                        </TD>
                        <TD>
                            <xsl:value-of select="@Birth_Year"/>
                        </TD>
                        <TD>

                        <xsl:choose>
                            <xsl:when test="@Death_Year[.='0']">
                            </xsl:when>
                            <xsl:otherwise>
                                <xsl:value-of select="@Death_Year"/>
                            </xsl:otherwise>
                        </xsl:choose>

                        </TD>
                        <TD>
                            <xsl:value-of select="@Title"/>
                        </TD>
                    </TR>
                </xsl:for-each>
            </TABLE>
            <br/>
    </body>
    </html>
</xsl:template>

</xsl:stylesheet>
```

4. Finally, I upload the generated XML file and corresponding XSLT into my personal website in virtual server.

The address is http://192.168.220.100/~andytang . And the shortcut to reach XML file I upload is http://192.168.220.100/~andytang/authordata.xml .

